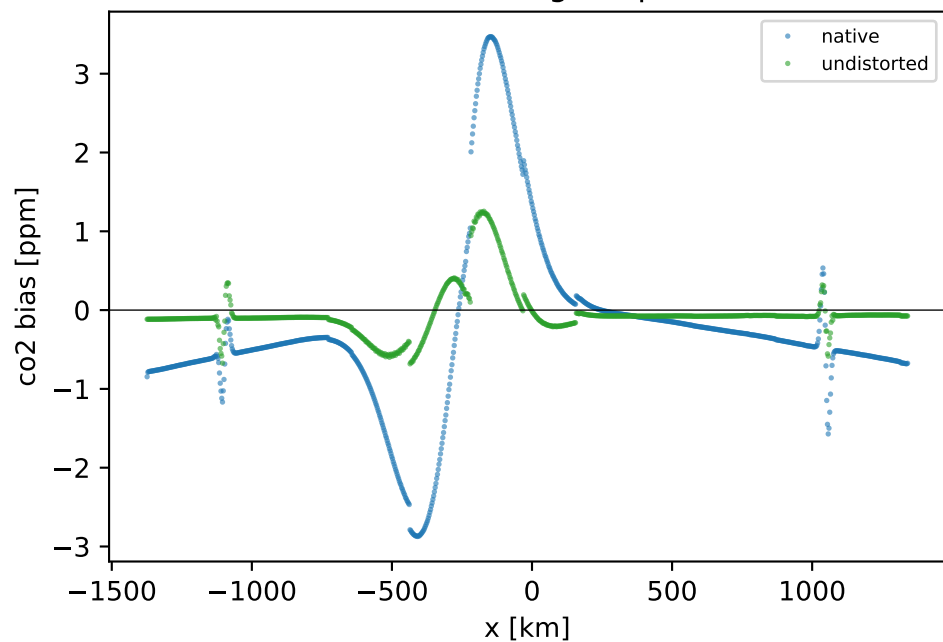
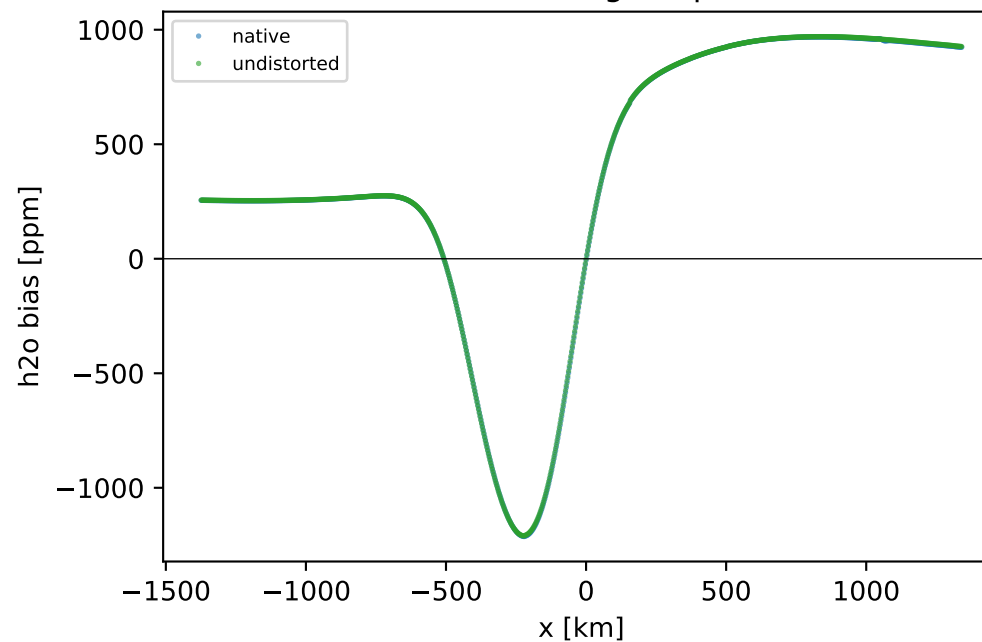


Joint FPA0+FPA1+FPA2 -- realistic, no noise (native/undistorted)

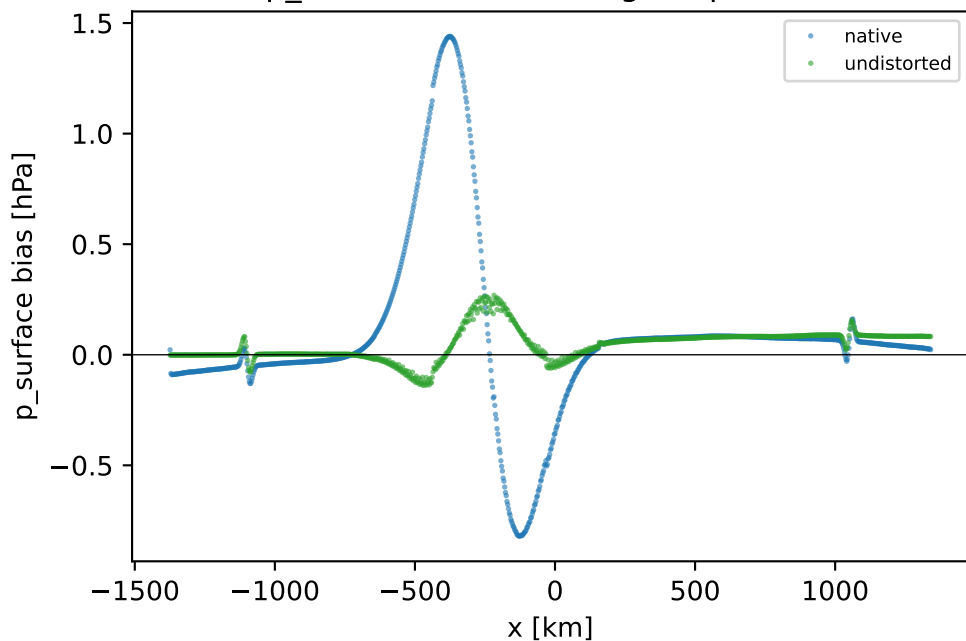
co2 bias vs along-slit position



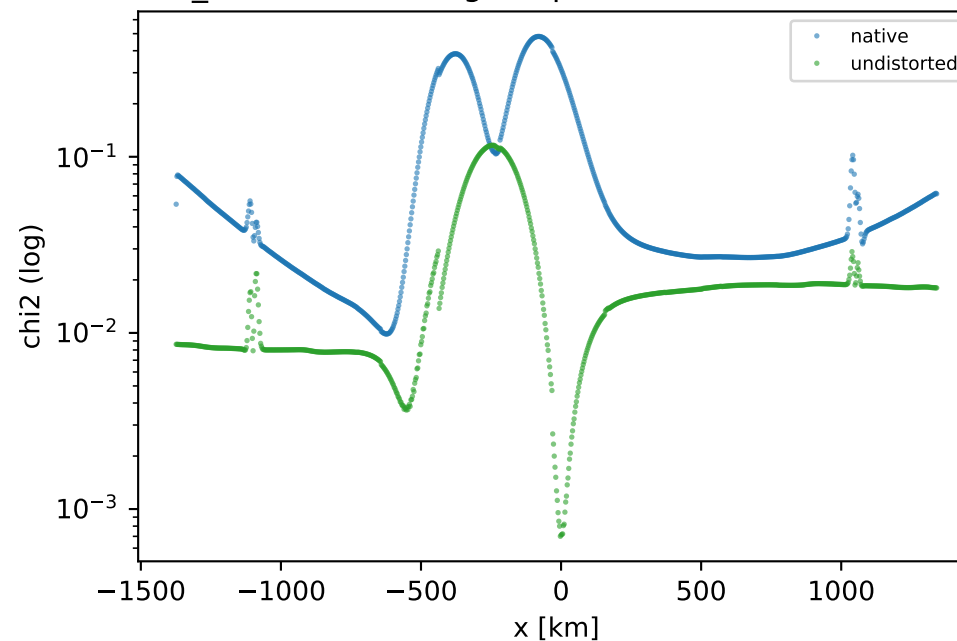
h2o bias vs along-slit position



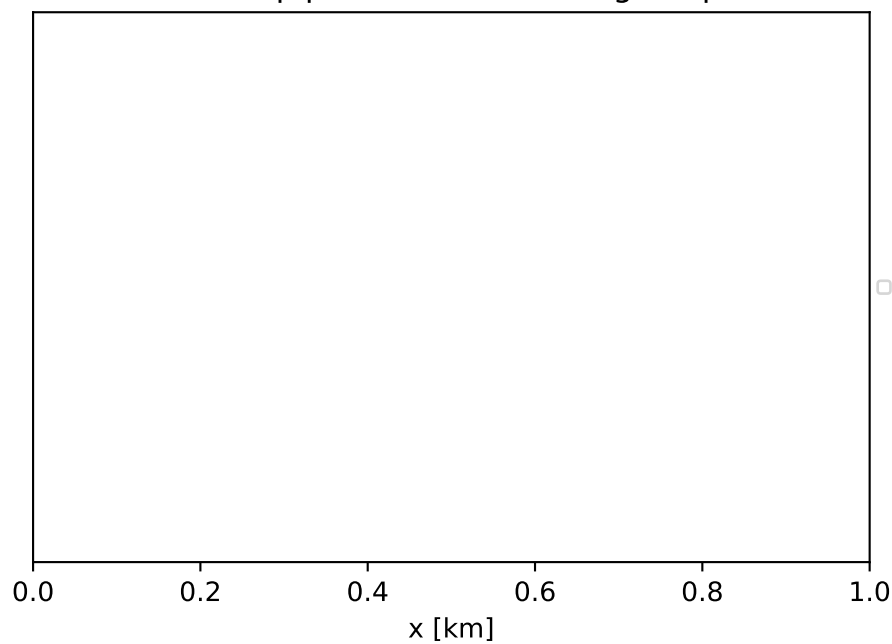
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

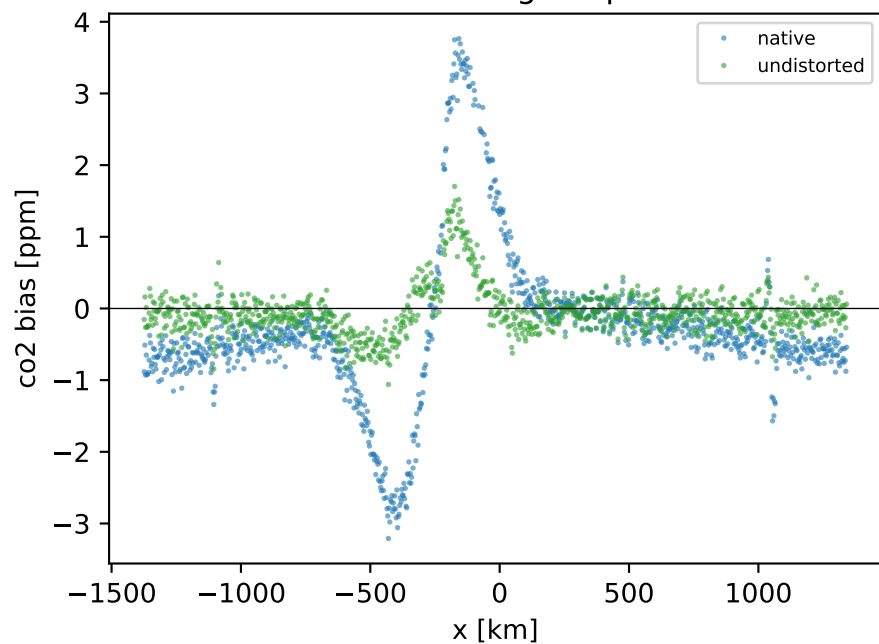


Where each pipeline fails, vs along-slit position

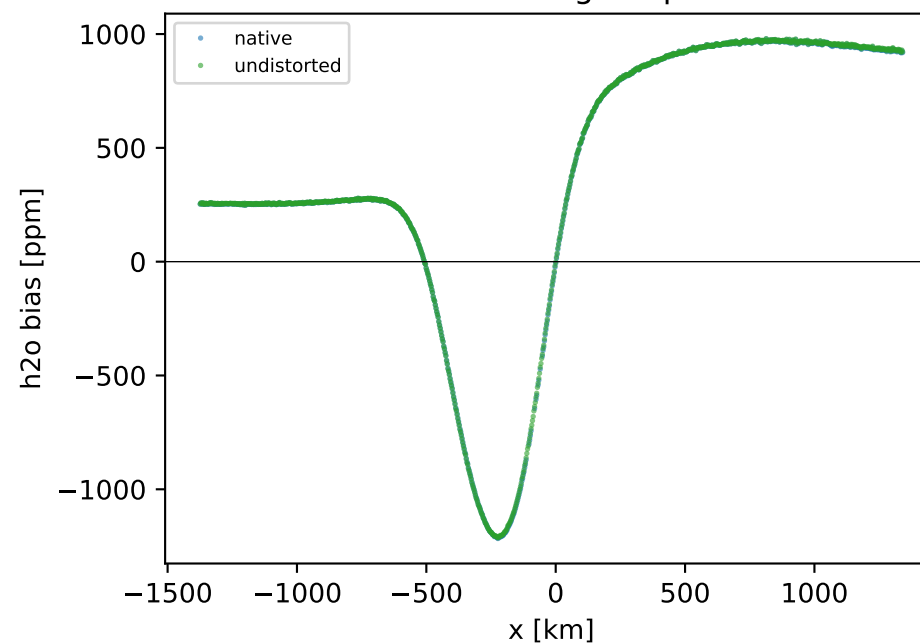


Joint FPA0+FPA1+FPA2 -- realistic, noise (native/undistorted)

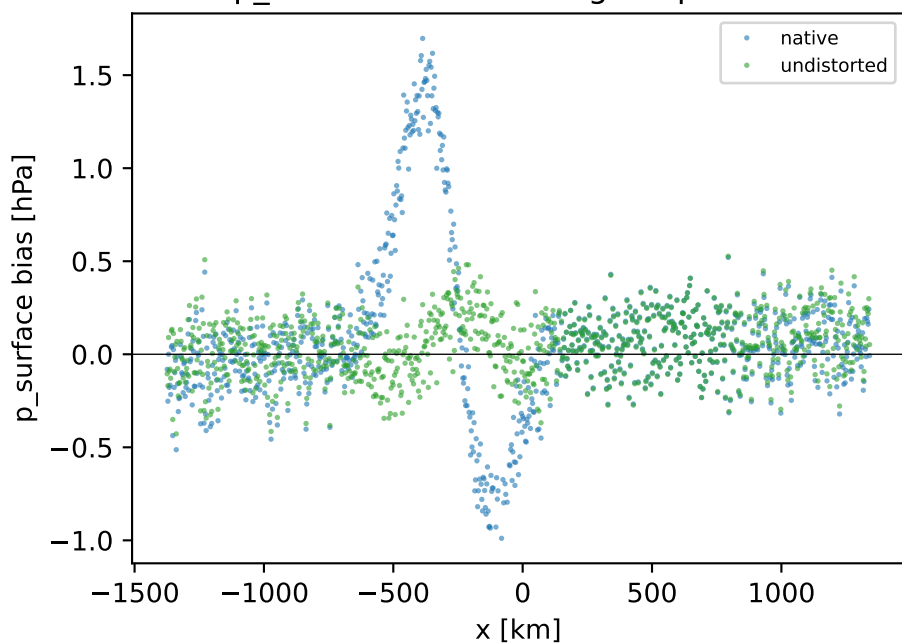
co2 bias vs along-slit position



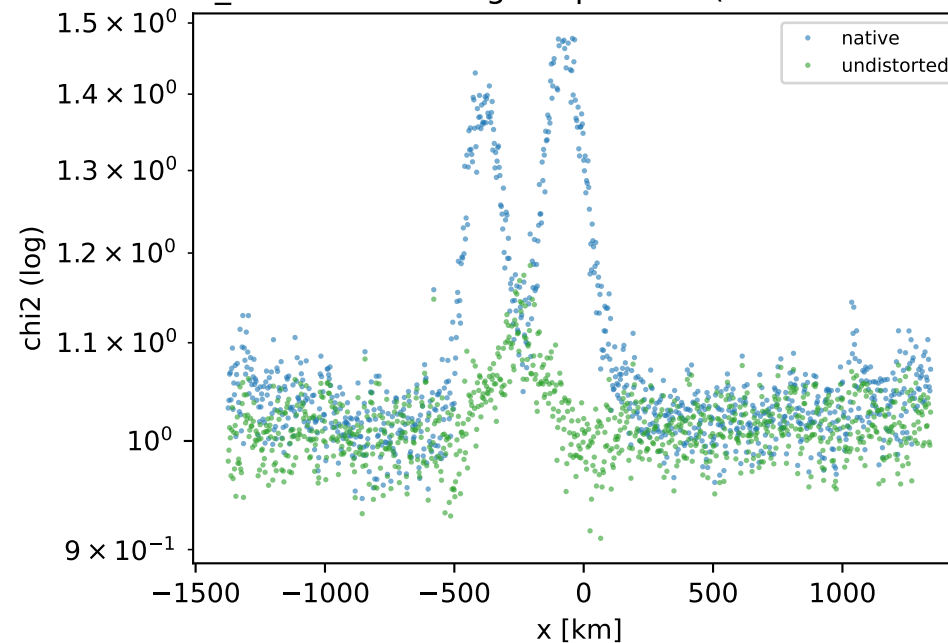
h2o bias vs along-slit position



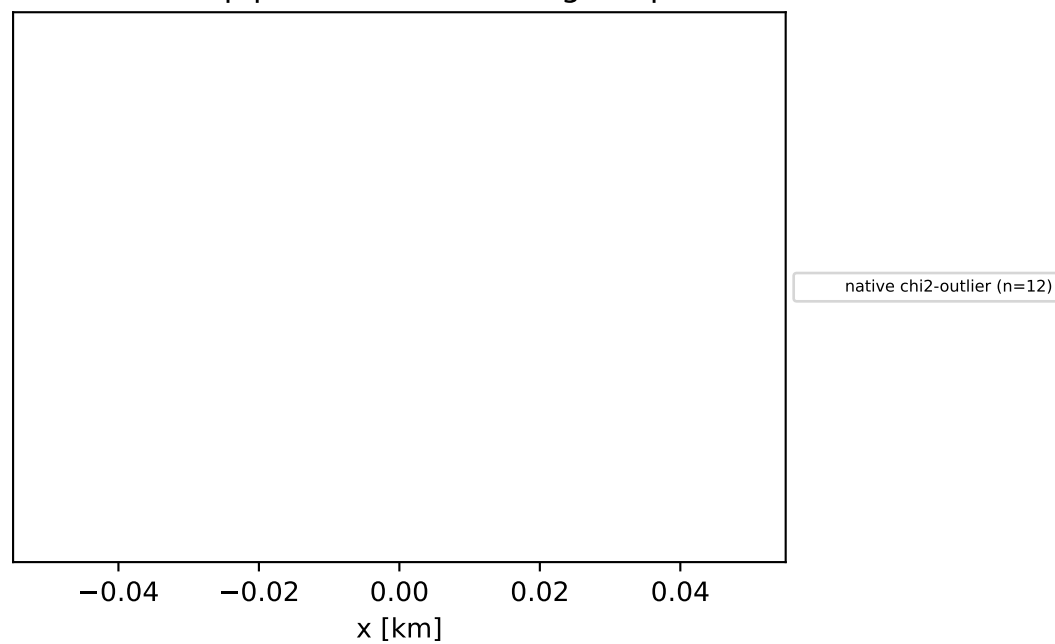
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

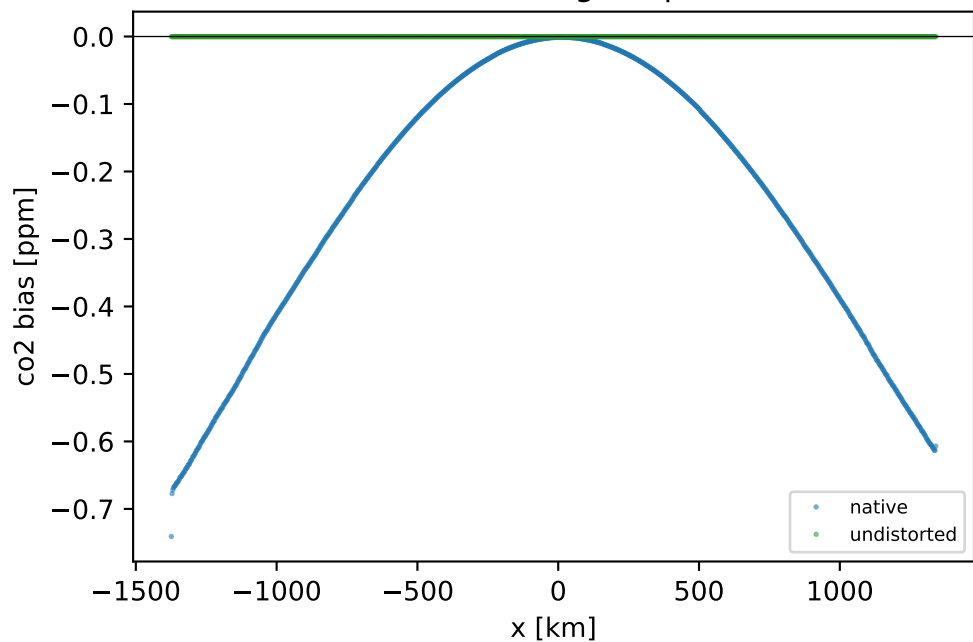


Where each pipeline fails, vs along-slit position

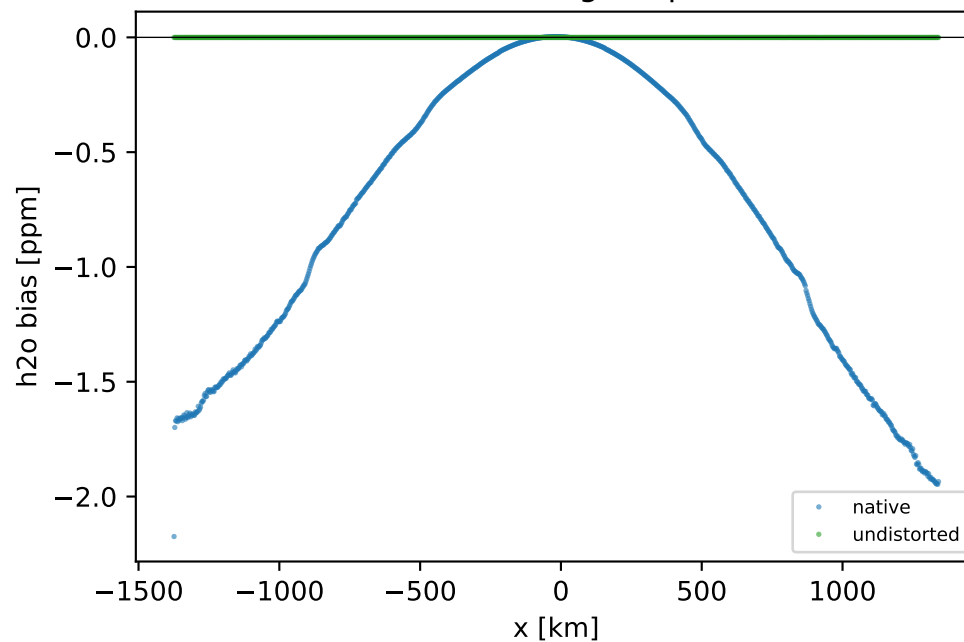


Joint FPA0+FPA1+FPA2 -- uniform, no noise (native/undistorted)

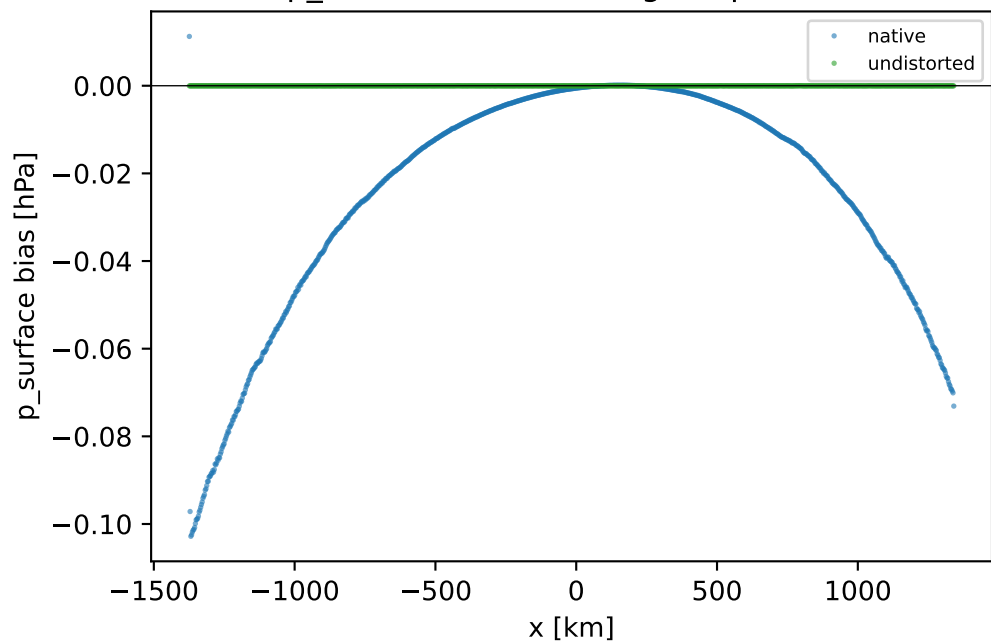
co2 bias vs along-slit position



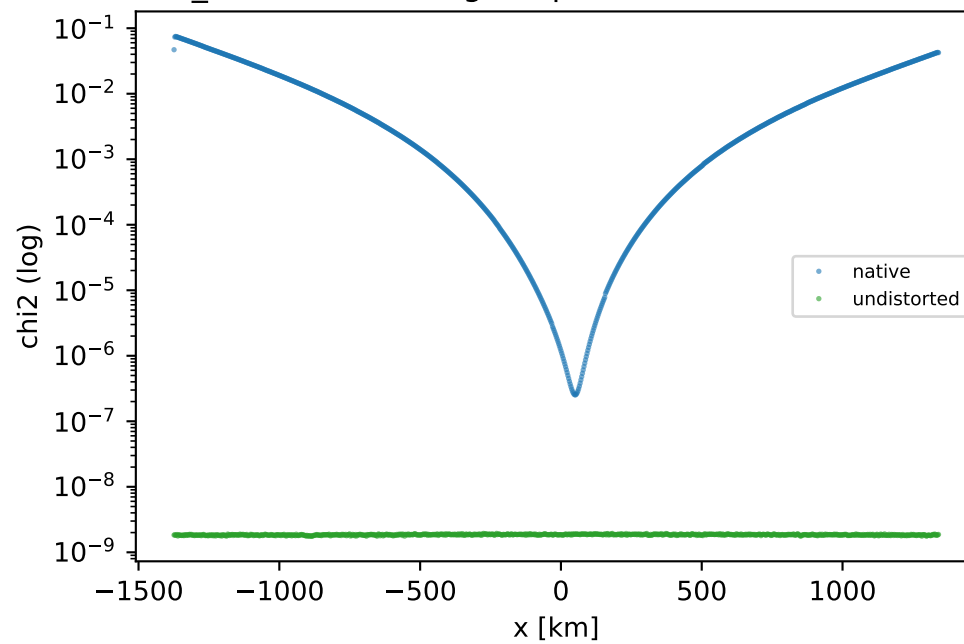
h2o bias vs along-slit position



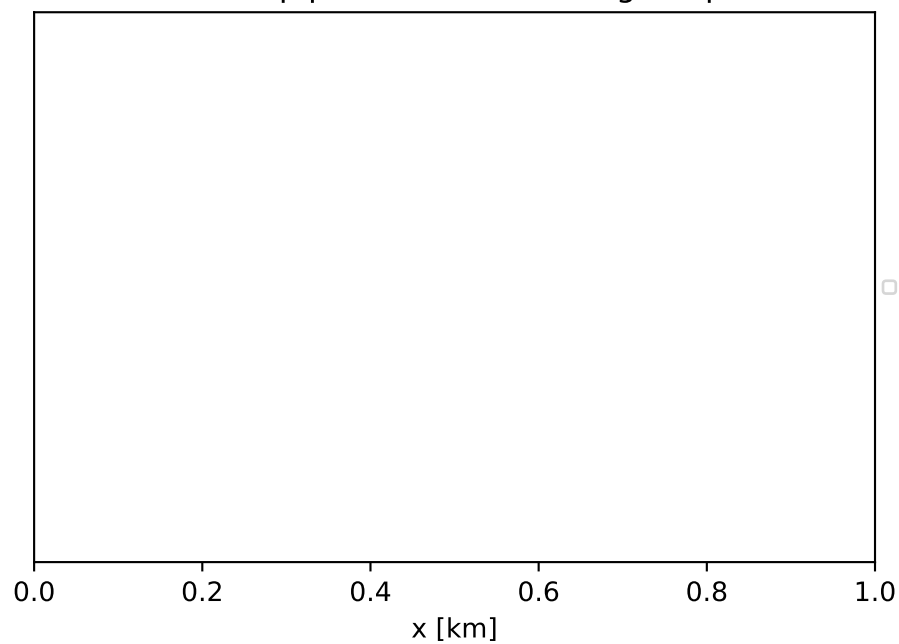
p_surface bias vs along-slit position



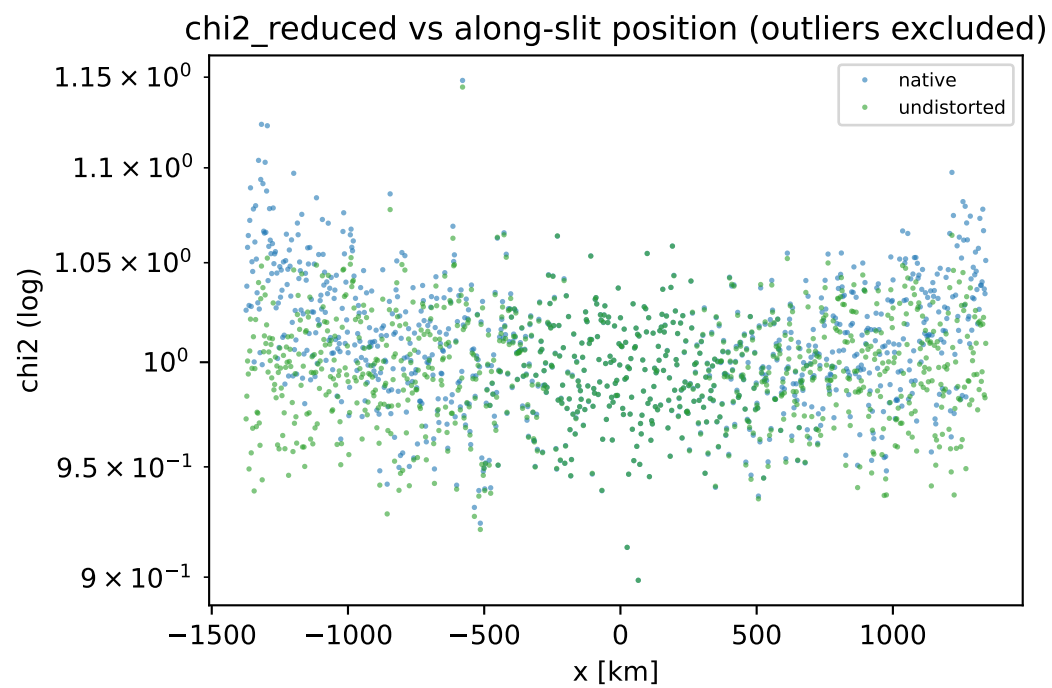
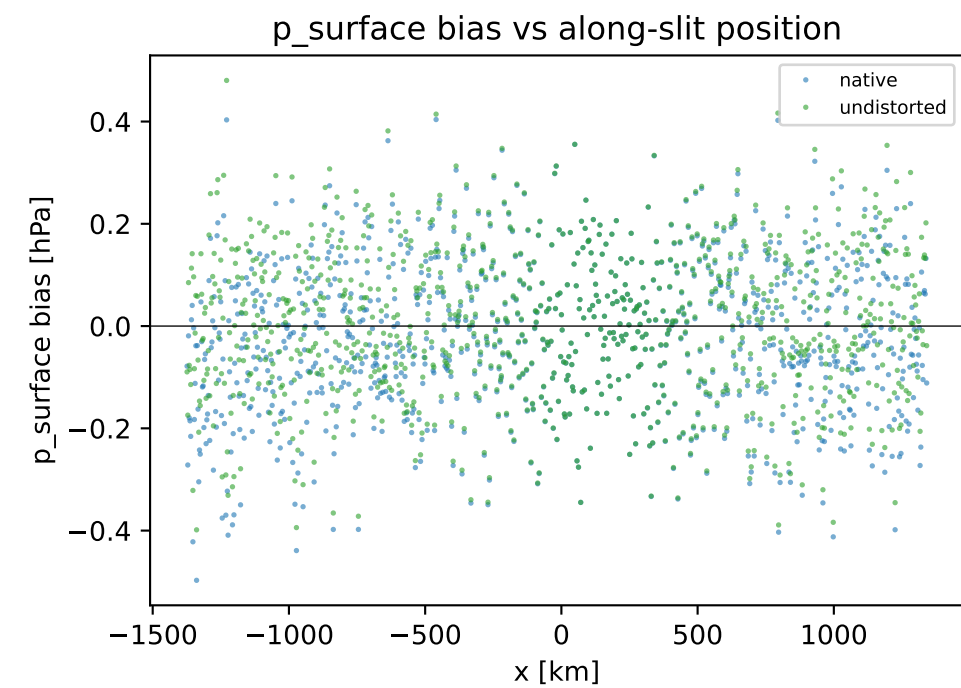
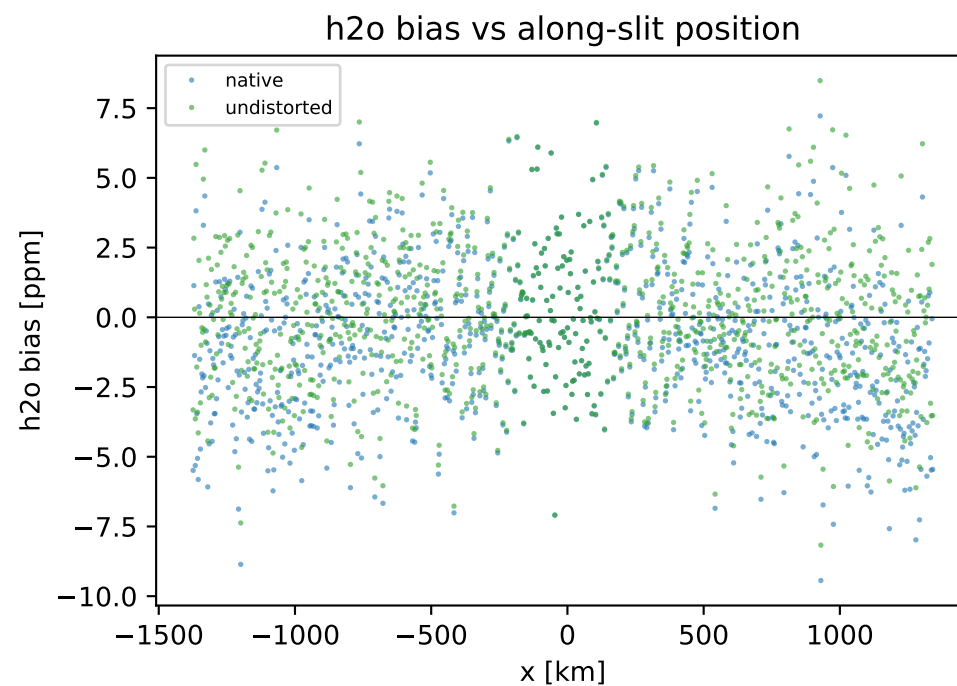
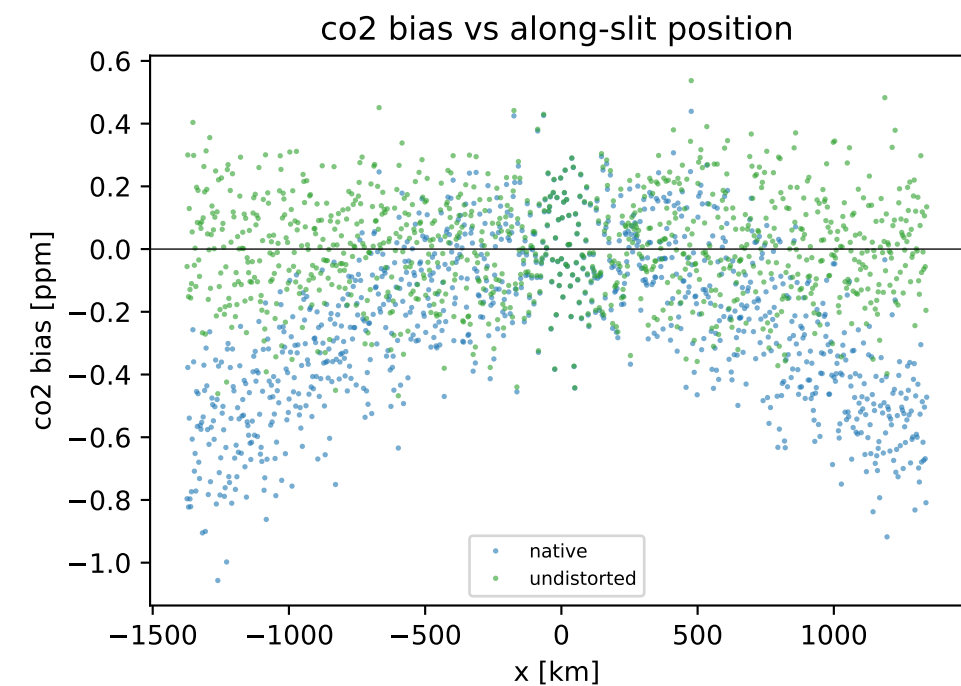
chi2_reduced vs along-slit position (outliers excluded)



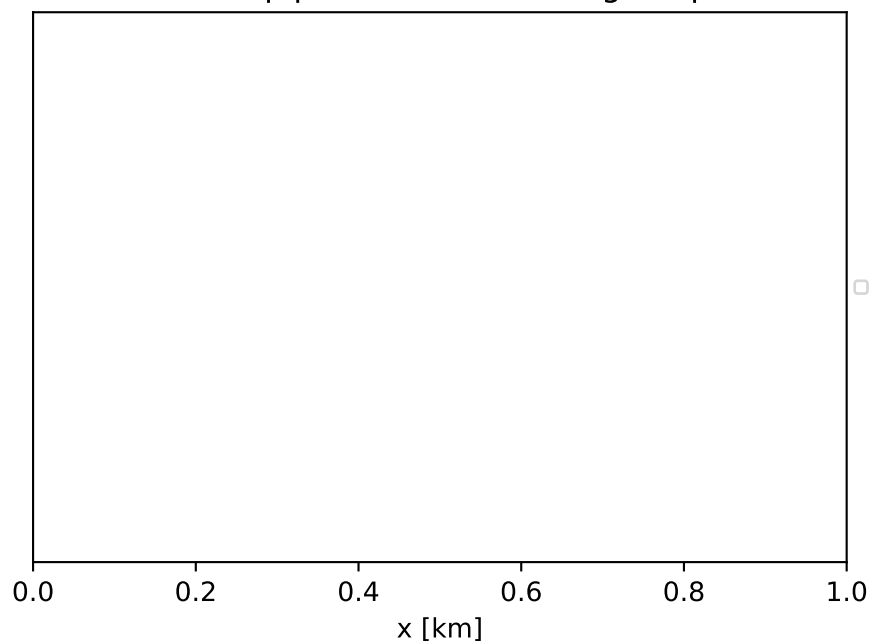
Where each pipeline fails, vs along-slit position



Joint FPA0+FPA1+FPA2 -- uniform, noise (native/undistorted)

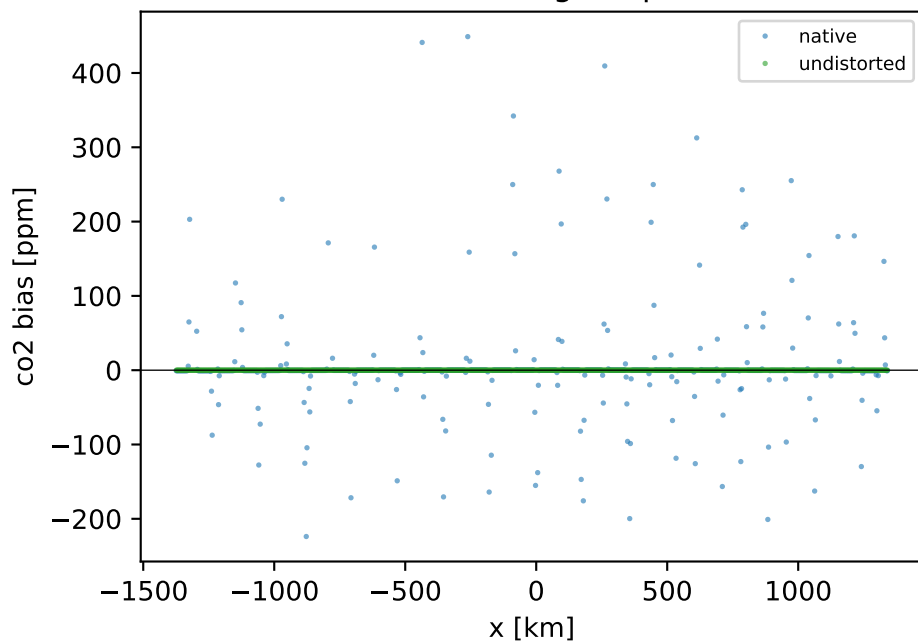


Where each pipeline fails, vs along-slit position

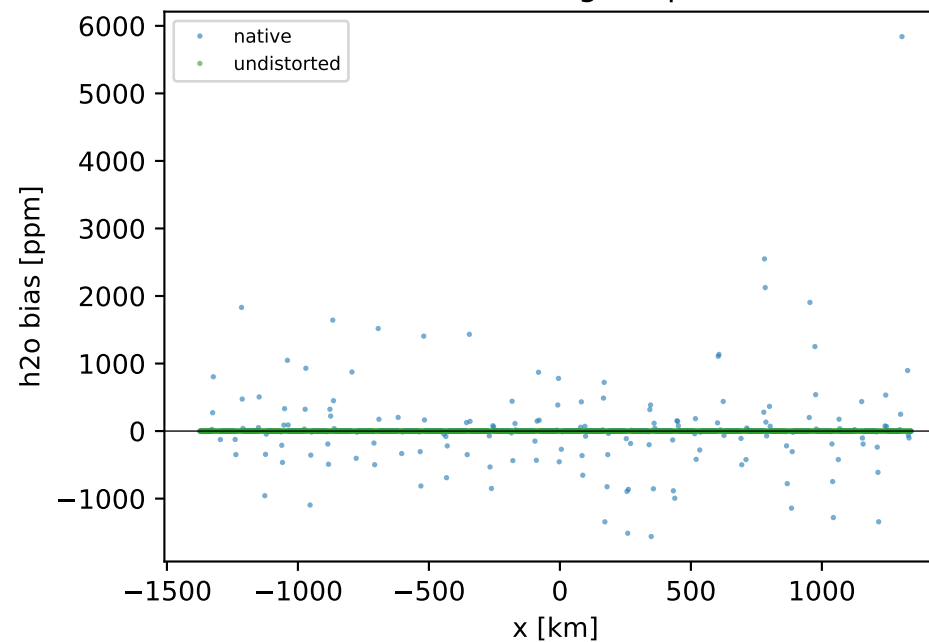


Joint FPA0+FPA1+FPA2 -- barcode, no noise (native/undistorted)

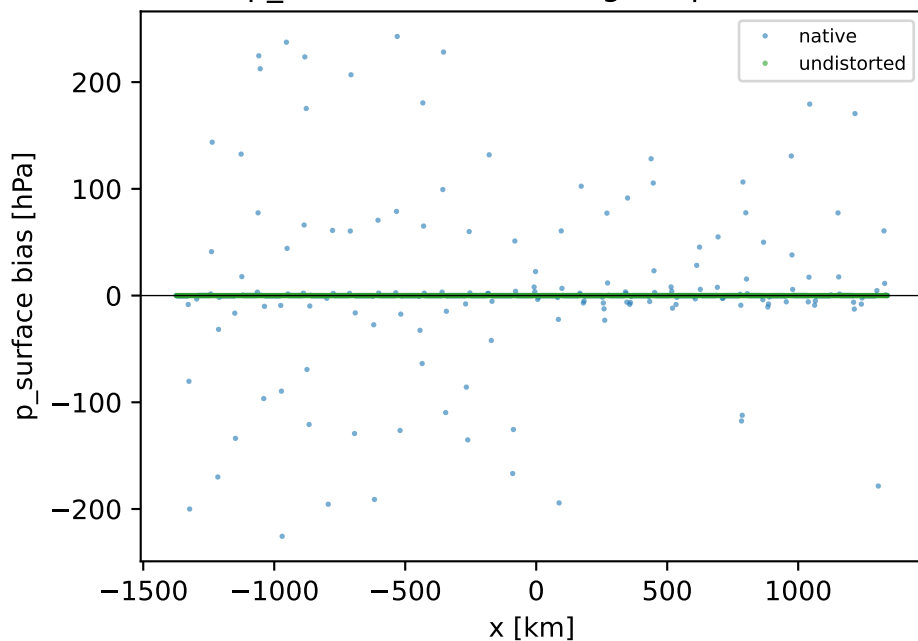
co2 bias vs along-slit position



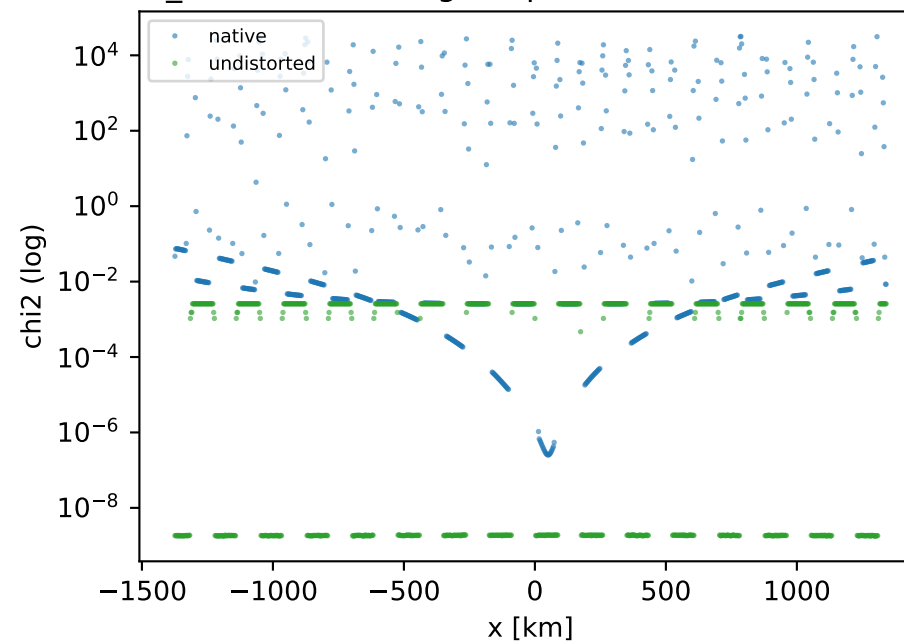
h2o bias vs along-slit position



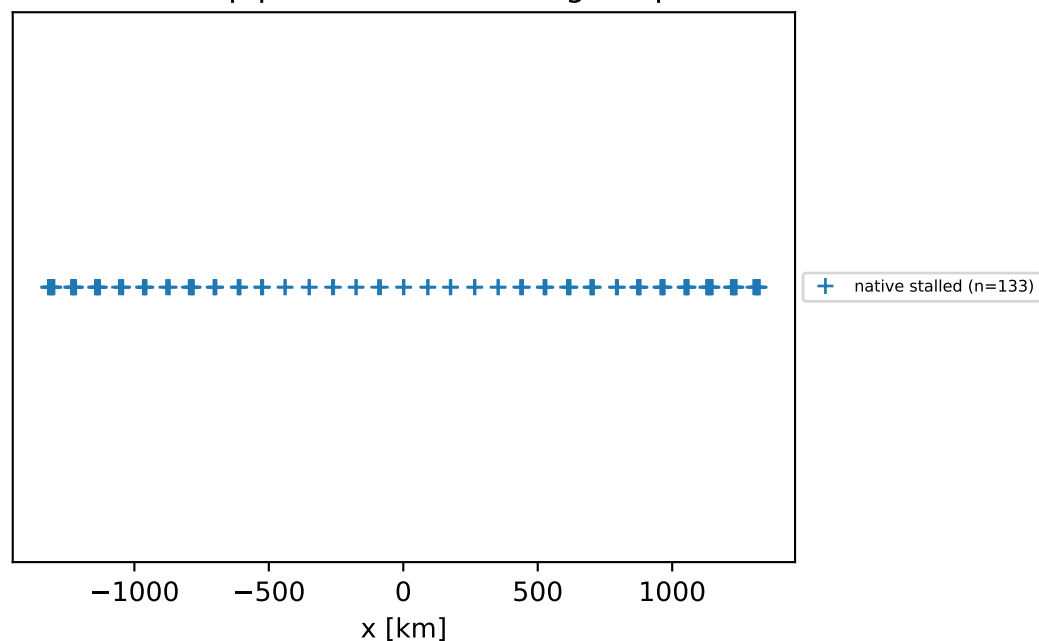
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

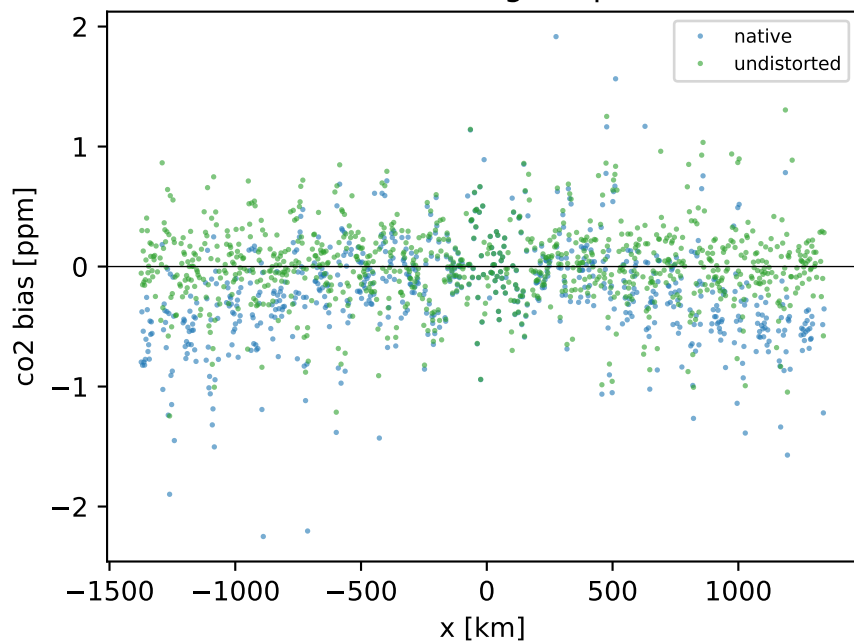


Where each pipeline fails, vs along-slit position

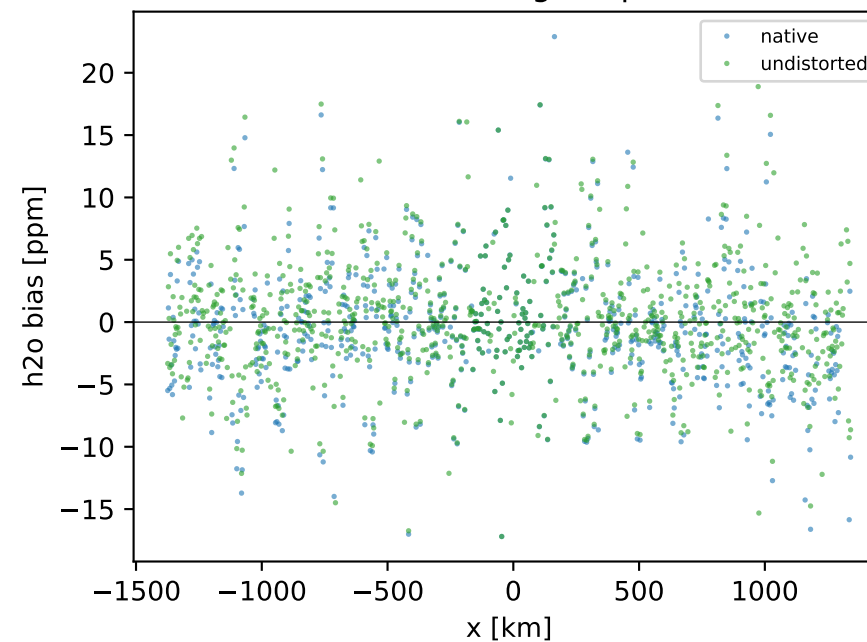


Joint FPA0+FPA1+FPA2 -- barcode, noise (native/undistorted)

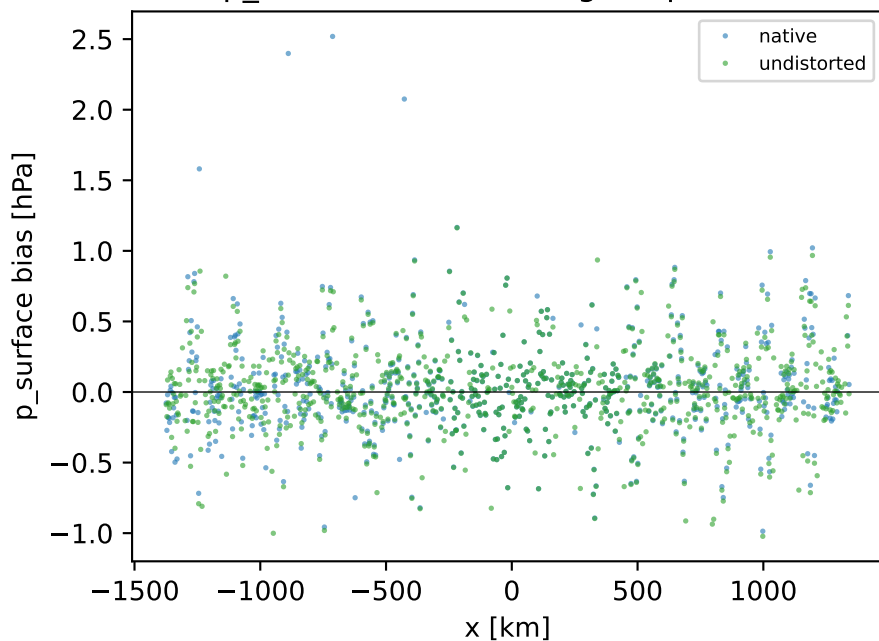
co2 bias vs along-slit position



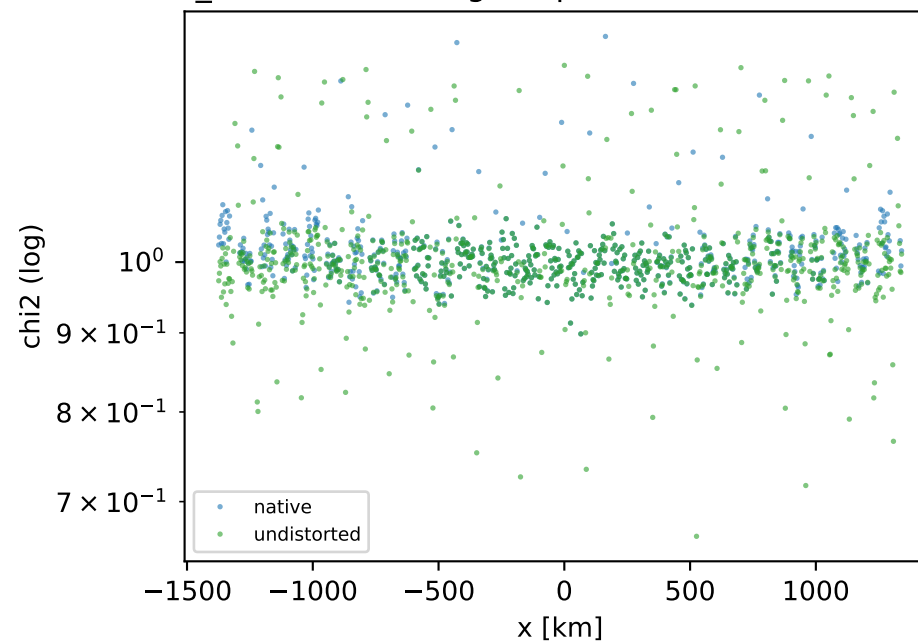
h2o bias vs along-slit position



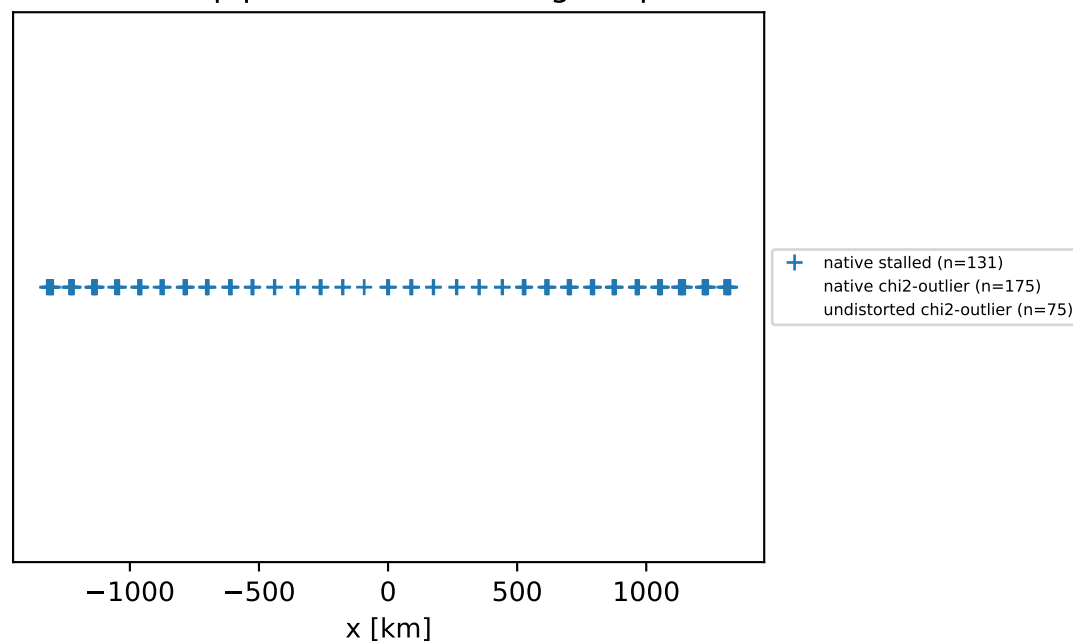
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)



Where each pipeline fails, vs along-slit position



FPA0+FPA1+FPA2 -- robust bias statistics across scenes/noise (native/undistorted)

scene	noise	quantity	pipeline	n	mean	std	median	MAD-sig	n_out
realistic	0	co2	native	1002	-0.2615	1.1	-0.3937	0.3844	44
realistic	0	h2o	native	1002	+367.3	628.7	+273.4	896.2	0
realistic	0	p_surface	native	1002	+0.08035	0.3924	+0.05234	0.09441	105
realistic	0	co2	undistorted	1002	-0.05099	0.2999	-0.07701	0.03372	199
realistic	0	h2o	undistorted	1002	+368.8	628.5	+273.9	896.5	0
realistic	0	p_surface	undistorted	1002	+0.04501	0.06923	+0.06037	0.07135	0
realistic	1	co2	native	990	-0.2918	1.069	-0.3999	0.435	26
realistic	1	h2o	native	990	+379.7	622	+274.3	884.2	0
realistic	1	p_surface	native	990	+0.09433	0.4153	+0.04578	0.2215	0
realistic	1	co2	undistorted	1002	-0.04554	0.3387	-0.08365	0.2235	0
realistic	1	h2o	undistorted	1002	+368.8	628.4	+274.2	897.6	0
realistic	1	p_surface	undistorted	1002	+0.04891	0.1624	+0.05215	0.1736	0
uniform	0	co2	native	1002	-0.2453	0.2025	-0.2043	0.2487	0
uniform	0	h2o	native	1002	-0.7739	0.606	-0.6877	0.8262	0
uniform	0	p_surface	native	1002	-0.02466	0.02587	-0.0148	0.02021	0
uniform	0	co2	undistorted	1002	-0.0001874	2.802e-06	-0.0001875	3.037e-06	0
uniform	0	h2o	undistorted	1002	-0.000493	3.728e-05	-0.0004938	3.816e-05	0
uniform	0	p_surface	undistorted	1002	-2.897e-05	7.353e-06	-3.071e-05	7.091e-06	0
uniform	1	co2	native	1002	-0.24	0.2614	-0.2222	0.2839	0
uniform	1	h2o	native	1002	-0.7308	2.586	-0.7509	2.535	0
uniform	1	p_surface	native	1002	-0.02078	0.1468	-0.02031	0.1506	0
uniform	1	co2	undistorted	1002	+0.005133	0.1657	+0.006247	0.1697	0
uniform	1	h2o	undistorted	1002	+0.04249	2.456	-0.04975	2.467	0
uniform	1	p_surface	undistorted	1002	+0.003875	0.1443	+0.007152	0.151	0
barcode	0	co2	native	869	+2.79	52.18	-0.1794	0.2594	157
barcode	0	h2o	native	869	+9.779	341.3	-0.5874	0.8674	174
barcode	0	p_surface	native	869	+1.532	36.82	-5.368e-05	0.03722	174
barcode	0	co2	undistorted	1002	-0.0001494	4.184e-05	-0.0001811	2.557e-05	0
barcode	0	h2o	undistorted	1002	-0.000323	0.0001895	-0.0004084	0.0002383	0
barcode	0	p_surface	undistorted	1002	-1.292e-06	2.938e-05	-8.522e-06	4.006e-05	0
barcode	1	co2	native	696	-0.22	0.4277	-0.2152	0.3569	0
barcode	1	h2o	native	696	-0.5003	4.93	-0.6216	3.567	0
barcode	1	p_surface	native	696	+0.03295	0.3443	-0.003356	0.2097	3
barcode	1	co2	undistorted	927	+0.007615	0.3236	+0.00266	0.2465	0
barcode	1	h2o	undistorted	927	+0.2352	4.739	-0.05955	3.467	0
barcode	1	p_surface	undistorted	927	+0.01622	0.2986	+0.007404	0.2144	0